

BuildAR: A Proof-of-Concept Prototype of Intelligent Augmented Reality in Construction

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Figure 1: A user experiencing the final BuildAR prototype.

ABSTRACT

Augmented Reality (AR) holds great potential to support the productivity, safety, and decision making of construction workers. We propose that AR in construction needs intelligent sensing, understanding, and adaptation based on context: information about the worker, their work/tasks, and the surrounding environment. As a proof-of-concept of intelligent AR, we present the design of the BuildAR prototype, which illustrates how a context-aware AR system might support an assembly task conducted by steel workers.

Index Terms: Human-centered computing—Human-Computer Interaction (HCI)—Interaction Paradigms—Mixed/Augmented Reality

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1 INTRODUCTION

Construction work sites are complex, hazard-prone, and relatively unstructured compared to other industry settings, leading to a variety of challenges for workers and managers. It is challenging for construction field workers to easily obtain relevant information, such as on-site training, warning about potential hazards, and instructions for conducting assigned activities. The challenge comes from multiple sources. First, accessing information in the field can put workers in a vulnerable situation for their safety. Second, the information needed to support construction workers can get complex, ranging from static information such as task instructions, to dynamic information about work-in-progress [7]. Lastly, the environment, work schedule, and plans change regularly. Such issues can place workers in hazardous situations, and can lead to reduced productivity and increased need for change orders, which may result in on the job injuries, project delays, and cost overruns [6].

To address these challenges, the required information conveyed to construction workers should be relevant, timely, and not cause distractions that can lead to safety hazards. We hypothesize that these challenges can be addressed through the use of augmented reality (AR). If every worker has a high-quality head-worn AR display, then information can be overlaid on the worker's view of the work site to provide instructions for unfamiliar tasks, warnings about safety hazards, and the latest construction updates. In addition, the pro-

vided on-the-spot instruction and training can prepare inexperienced workers in a more effective way and in a shorter amount of time. This can further help in addressing the workforce shortage crisis the industry is currently facing [5].

Recognizing the potential benefits of AR in construction, significant prior work has been done to prototype AR-based applications in this domain. For example, in Zhou et al.'s work [9], an AR-based system was used to inspect segment displacement during tunneling construction on-site in real-time, and most of the interviewees believed the system to be more effective than the conventional method. On-site information retrieval has been a longstanding problem within construction work sites. Yeh et al. constructed a low-fidelity AR helmet, iHelmet [8], using a projector and iPod touch to combat this issue, which resulted in users being able to retrieve information more efficiently and accurately when using the device over traditional 2D drawings. Garbet et al. [1] implemented marker-based AR connected to a cloud-based database, to provide the ability to view, interact and collaborate with 3D and 2D BIM data via AR for geographically dispersed teams. Harikrishnan et al. [3] also investigated the communication scenarios in which AR may be able to address construction-related communication needs through semi-structured interviews. The findings showed that practitioners perceived AR to be of value in improving the communication between the design team and construction team; leveraging the expertise of experienced personnel; and aiding in inspections.

However, we argue that AR will not reach its full potential for construction until AR systems are context-aware and intelligent. Simply having an AR display and a database of information about schedules, drawings, tasks, and the like is insufficient, because we cannot expect construction workers to interact manually with the AR system to search for the right information to display in every situation. We envision an *intelligent AR* system that can sense and understand the context (worker, work/task, and environment) and intelligently adapt the display to present the right information, at the right time, and in the right form for the worker's current needs.

In this paper, we introduce the concept of intelligent AR (section 2), and present a prototype called BuildAR that illustrates how that concept might be applied to a construction task (section 3). The prototype illuminates both the potential and challenges of intelligent AR, and leads to a research agenda (section 4) for the future.

2 INTELLIGENT AUGMENTED REALITY

With high-quality AR glasses, users can work anywhere, and can see documents or images at full size even while on the go. The possible applications are practically limitless. However, with this extreme power and flexibility, challenges also arise. When users can display any amount of information at any time, how do we avoid information overload, distraction, and safety hazards? When apps are always present, how do we keep them unobtrusive? Clearly, we must somehow limit the information visible to the user, and limit negative effects on tasks in the physical world, while still achieving the goals of pervasive AR [2]. The key question is: *how can we enable AR users to access the right information at the right time without compromising real-world activities, and without requiring excessive manual interaction?* This question suggests that the AR system must be intelligent—the system should be able to predict what information is needed, and when/where/how to display it. For the system to be able to do this, it must be able to understand the user's *context*: where are they, what are they looking at, what objects are present, what are their goals, what information do they need, what is their current cognitive state, and what is their history and preference?

We envision intelligent AR systems that can detect, understand, and reason about their context of use, and that can adapt their displays and interactions based on that context. "Context" refers not only to the environment and objects around the user, but also knowl-

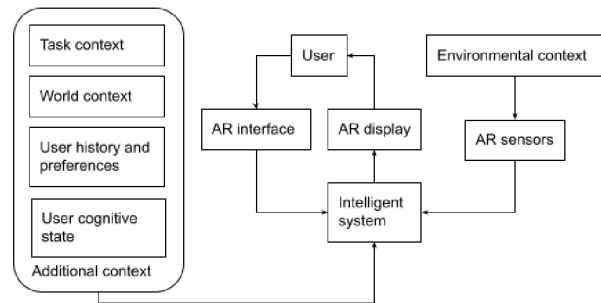


Figure 2: Intelligent AR system concept.

edge about the task, the world, and the user themselves—their history, preferences, and cognitive state (e.g., workload, fatigue). This concept is illustrated in Figure 2.

To realize this vision requires significant advances not only in artificial intelligence (e.g., machine learning, computer vision), but also in AR information display and interaction techniques, as well as in intelligent user interfaces.

3 BUILDAR

To begin exploring the space of intelligent AR in construction, we designed and prototyped a proof-of-concept system called BuildAR.

3.1 Project Goals

BuildAR is intended to be an intelligent AR system that guides construction workers in completing tasks on site with ease and safety. The goal of this project was to design a working, interactive proof-of-concept to illustrate the idea of intelligent AR described in the previous section. We wanted to be able to visualize the concept and to imagine other possible use cases. We also aimed to further understand the potential challenges and benefits of intelligent AR in an authentic construction use case. We hoped that the project would give us a better understanding of potential future applications of intelligent AR.

In terms of the system's design, our goal was to create an intuitive seamless interface that would be a powerful asset on the construction site. In the construction industry, workers are faced with daily obstacles and challenges that are mentally and physically demanding. By alleviating some of the mental pressure through perceptual and cognitive enhancements provided by AR, we hoped to enable workers to be safer and more productive. We designed all features of BuildAR to follow these design guidelines.

3.2 Data Gathering and Analysis

We began our design process with contextual inquiry and contextual analysis to gather and organize information about the construction domain and its challenges. In contextual inquiry, we gathered information relevant to the system design from a variety of sources, while in contextual analysis, we represented the gathered information as "work activity notes" and organized them using work activity affinity diagramming (WAAD) [4].

The first part of our inquiry was to visit a construction site to get an initial understanding of the type of work and workers that are involved in a construction project. We arranged a meeting with a site supervisor to give us information on the workflow of an ongoing construction project in Blacksburg, Virginia. During the visit, we asked questions and gathered data to inform us about the system we were going to design. We discussed the nature of construction jobs and roles, crucial construction topics to research, the purpose of the system, and possible features. The supervisor emphasized

that an AR system should not block the view of any worker who used it, especially if that worker performed their job in precarious positions. Our site visit also helped us understand how different workers would need different features, which points to the need for personalization and adaptivity. Another takeaway that informed our initial design ideas was that design-build construction projects often require many submittals and requests for information as the project changes during construction.

We also gathered information from an academic expert in building construction (one of the co-authors of this paper). The expert reemphasized that the system needs to be safe for workers to use on site. We also learned that a major problem on construction sites is that workers often don't have the most up-to-date information as soon as they need it. Most importantly, the expert helped us narrow the scope of our project to focus on individual worker profiles. To flesh out this aspect of our information gathering, we conducted internet research on construction site roles.

As part of our contextual inquiry, each member of our team took part in an experiment looking at the effectiveness of verbal instructions versus paper drawings on completing a faux steel beam connection. The experiment had us complete a connection of 3D printed parts using nuts and bolts, first, using automated verbal instructions, then using paper schematics of the connection. This experiment helped us get a much better understanding of the type of work steelworkers do on site. The experiment gave us insight into how we might use AR to aid workers who may be less experienced with their task. This was crucial in informing the design of the BuildAR system.

Additionally, we crafted two personas while designing the BuildAR system. The first persona was a construction site supervisor meant to represent site management. Our second persona was an inexperienced steelworker that was designed to be an example end user of our final prototype of BuildAR.

3.3 Design Ideas

Using rapid ideation and critique, we brainstormed a number of design ideas and features for BuildAR.

We initially wanted to focus BuildAR on construction management. The first idea was to allow the user to view the most recent construction models overlaid on the construction site in order to survey the progress of the project. We thought that annotating these models would allow the user to give live information about where pieces needed to be placed or moved. The system design then shifted towards allowing the user to manipulate the construction models using hand gestures. This was to allow the user to virtually place the annotated model somewhere in the real world and rotate and scale it to help them work on the model. However, these ideas were eventually viewed as too conventional, and not truly representative of the intelligent AR concept. To incorporate more intelligence into the system, we had some ideas about smart scheduling or layered models with adaptive changes when one layer was changed, but these ideas were either too broad in scope or not feasible to prototype.

We then shifted the focus of BuildAR to be for inexperienced construction workers like novice steelworkers. Our system design became centered on helping workers with their task by showing them 3D models to help walk them through their task step by step. We designed a system that allows the user to first select a task to complete out of their assignments. Once the user selects the task, they are given the list of materials they will need to complete the given task. Once the user has gathered all the necessary materials, BuildAR will show annotated models for each step of the process of completing the task. With artificial intelligence, BuildAR will be able to determine where the user is located relative to their task location and when the user is finished with their current step. Once the user has completed a step, the system automatically advances them to the next one.



Figure 3: The 3D printed model of a steel beam connection used in the final prototype (top) and the AR model of the part shown to the user to aid in assembly (bottom).

Because construction sites can be very dangerous, we needed to make sure that our system would not be too distracting for the user. Thus, we also wanted to come up with a way for BuildAR to help the user stay safe by always being aware of their surroundings. We originally considered allowing the user to take a picture of a hazard on the job site and upload it so other employees can be aware of hazards near them. However, we decided that BuildAR should know where the user and potential hazards are located on the job site and alert the user when they are getting close to a hazard. This idea would use computer vision to be able to recognize unsafe conditions and alert the user when they are near a hazard.

We also came up with some ideas to help the user stay in contact with other employees and to keep up with the updated construction schedule. Communication on the job site could be improved by BuildAR alerting the user when they receive an email, phone call, or text message. We also had the idea to allow the user to view the most recent construction schedule and adjust the schedule according to any delays that may occur. However, these communication and notification ideas were not fully realized in our prototype.

3.4 Final Prototype

The final prototype of the BuildAR system was a working, interactive AR demo of our system design on the Microsoft HoloLens 2. The HoloLens 2 application was developed in Unity 2019.4.26f1 using the Microsoft Mixed Reality Toolkit. Along with the AR aspect of the demo, we used a physical 3D printed model of a steel beam connection that the user completes with the guidance of BuildAR by referring to a 3D virtual model of the connection (Figure 3).

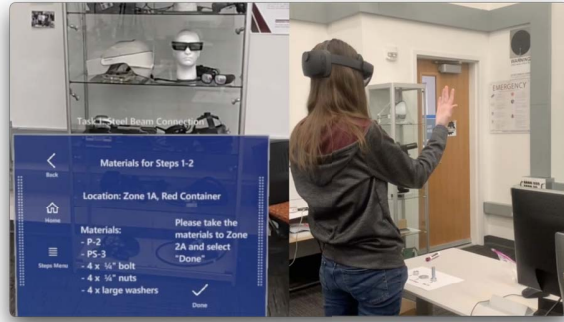


Figure 4: AR instructions guiding the user to materials needed for the connection task.

The prototype demonstrates five envisioned intelligent features:

- User recognition: the construction worker is automatically recognized when he/she dons the AR headset.
- Knowledge of user tasks: the system knows what tasks have been assigned to the recognized user.
- Guidance to materials' location: the system knows what materials are needed for the selected task and where they are located, so that it can guide the user to find them (Figure 4).
- Hazard detection and warning: the system automatically recognizes hazards in the environment and marks them in AR to warn the user.
- Detection of task step completion: the system detects when each step of the task is completed, and displays the next step and its associated 3D model automatically.

Because the focus of our work was on system design and prototyping, we implemented these features either by hard-coding values into the application, or by a Wizard-of-Oz approach in which one of the demo staff uses a keyboard to "intelligently" change the system state based on user actions.

The demo experience involves an oral introduction to our project as well as some background information the user needs to know to use the application. After the introduction, the user can then navigate through menus to select the task of a steel beam connection. The user has to collect the materials indicated by the system and walk to the area where they will be completing the connection. Between the user and the connection area is a labeled hazard that users need to circumnavigate. Once the user is in the correct place to start the task, we use a wireless keyboard to advance the user to the first step of connecting the pieces. The wireless triggering of advancement is meant to imitate the intelligent part of the system. At each step of connection, the user has a virtual 3D model that they can move, rotate, and scale of the connection they need to make for the respective step (Figure 3, bottom). Once the user has completed the connection, a pop-up dialogue automatically appears, telling them they have completed the demo.

4 RESEARCH AGENDA

The convergence of several maturing technologies is poised to accelerate how we work. Notably technologies such as AR, artificial intelligence (AI)/machine learning (ML), computer vision (CV), Internet of Things (IoT), and wearable sensors all offer unique opportunities to fully understand workers' context and provide time-critical,

real-time information to increase productivity and safety. This is especially useful in dynamic complex occupational settings, where disparate teams are still using paper, or manually synchronized files viewed on tablets and phones. However, these technologies need fundamental research advancements in order to truly impact the future of work. Thus, our initial work here aims to set forth a research agenda for a number of convergent research thrusts, each mature in its own right, and ripe for integration into Intelligent AR applications.

For example, work is needed in AI/ML and CV to advance the state-of-the art in real-time understanding of user's work context (such as the automatic recognition of task completion in the BuildAR prototype). This includes efforts to develop approaches to organize the multiple perception tasks (action recognition, object detection, location recognition, task state tracking, and many more), develop models that can correctly integrate multisource information (e.g., from multiple sensors and locations), and develop models that can self-train and self-update to continuously improve real-time prediction accuracy. Additionally, techniques are needed to transfer the learned knowledge/models from a given task or target domain to another, so that once the system understands a construction-related task such as beam assembly, the model can be slightly altered to scale to other tasks and domains, such as part assembly in manufacturing.

Other research inquiries lie closer to the worker experience afforded by Intelligent AR user interfaces. These include identifying novel approaches to ensure AR users can seamlessly access the right information at the right time without compromising their real-world work activities, and without requiring excessive manual interaction, as well as developing approaches to automatically arrange AR information during work so that it's easy to see and also unobtrusive (in BuildAR, we found that manually placing the 3D model and the task instructions in our prototype was time consuming and distracted users from the task). Work is further needed to develop and validate new guidelines for automating the display of AR information during work so that users can minimize explicit interactions with the system (e.g., in BuildAR, the system can automatically display the list of tasks because it recognizes the user's identity). Lastly, intelligently (and automatically) adapting to the worker's state while they are working is a rich space for innovation, where, for example, new techniques are needed to adjust AR information presentation when workers are fatigued, slower "than usual" in task performance, exhibit noisy manual responses and/or are making errors, and can identify and prioritize contextual elements across different Intelligent AR scenarios as a worker's day progresses.

5 CONCLUSIONS

Our intelligent AR work envisions how the convergence of AI, AR, and related technologies can have a significant positive impact on the future of work in building construction and other complex, unstructured domains. The BuildAR proof-of-concept demonstrates that this approach can be a powerful way to support construction workers, but that significant technical and design-oriented challenges must be addressed. Our interdisciplinary team plans to build upon this work by tackling the proposed research agenda to achieve intelligent AR's potential in the coming years.

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REFERENCES

- [1] J. Garbett, T. Hartley, and D. Heesom. A multi-user collaborative bim-ar system to support design and construction. *Automation in Construction*, 122:103487, 2021.

- [2] J. Grubert, T. Langlotz, S. Zollmann, and H. Regenbrecht. Towards pervasive augmented reality: Context-awareness in augmented reality. *IEEE transactions on visualization and computer graphics*, 23(6):1706–1724, 2016.
- [3] A. Harikrishnan, A. S. Abdallah, S. K. Ayer, M. El Asmar, and P. Tang. Feasibility of augmented reality technology for communication in the construction industry. *Advanced Engineering Informatics*, 50:101363, 2021.
- [4] R. Hartson and P. Pyla. *The UX book: Agile UX design for a quality user experience*. Morgan Kaufmann, 2018.
- [5] S. Kim, S. Chang, and D. Castro-Lacouture. Dynamic modeling for analyzing impacts of skilled labor shortage on construction project management. *Journal of Management in Engineering*, 36(1):04019035, 2020.
- [6] . Nedeljković and M. Kovačević. Building a construction project keyphrase network from unstructured text documents. *Journal of Computing in Civil Engineering*, 31(6):04017058, 2017.
- [7] D. Tian, M. Li, J. Shi, Y. Shen, and S. Han. On-site text classification and knowledge mining for large-scale projects construction by integrated intelligent approach. *Advanced Engineering Informatics*, 49:101355, 2021.
- [8] K.-C. Yeh, M.-H. Tsai, and S.-C. Kang. On-site building information retrieval by using projection-based augmented reality. *Journal of Computing in Civil Engineering*, 26(3):342–355, 2012.
- [9] Y. Zhou, H. Luo, and Y. Yang. Implementation of augmented reality for segment displacement inspection during tunneling construction. *Automation in Construction*, 82:112–121, 2017.